

Project Plan

Data Analytics Capstone || MGMT 6219

Riipen Industry Project: Bumby Wool



Submitted by:

Baranitharan Balraju

Marvelous Chidiebube Ihedigbo

Rahul

Arun Saji

Antony Boon Clinu Mangatt Raju

Submitted to:

Ellen Husk

Problem Statement

Bumby Wool is a Canadian-made, sustainable fashion brand that creates highly customizable wool outfits via an e-commerce platform. The company offers and promotes products on three platforms: Shopify (transactions and product data), Meta/Facebook and Instagram (paid and organic marketing) and Google Analytics (website behaviour and traffic). As Bumby Wool has expanded, so has the amount of information generated across these platforms, but the company currently has no working centralised, structured process to convert this data into insightful action.

The common problem is fragmentation: sales, marketing and website performance are recorded in different ways, are not consistently reconciling them, and do not have a defined set of key performance indicators (KPIs). Consequently, it can be difficult for the founder to determine which products or campaigns are helping to grow, where customers are leaving before buying, and what type of marketing is creating a positive ROI.

We will evaluate the current data collection and usage practices of Bumby Wool, develop a KPI roadmap in line with the business' growth objectives, and create a feasible, cost-effective solution to connect Shopify, Meta and Google Analytics data within a unified reporting framework. The analysis of available data will be used to identify trends and opportunities, create a dashboard blueprint that a small team can work on in a realistic time frame and present it along with a report and actionable recommendations for marketing, product and operations decisions.

Research Questions

Here are the questions we're aiming to answer:

- Which KPIs give the clearest picture of how Bumby Wool is really doing in sales, marketing, and keeping customers?
- Which products (and which types of customization) bring in the most revenue and profit?
- What's a realistic way to bring Shopify, Meta, and Google Analytics data together, given our tools and timeline?
- Out of everyone who visits the website, how many actually buy something and where do most people drop off along the way?
- Is the money spent on Meta ads actually bringing in profitable customers (looking at ROAS and cost per acquisition)?
- Do sales, traffic, or marketing results change by month or season?
- What kind of charts or dashboard views would actually help the founder make faster decisions week to week?

Connection with Other Courses

This project ties into a few other courses we're taking:

- Predictive Analytics - we'll look at past sales and traffic patterns to help predict what's coming next, like busy seasons.
- Data Strategy - we'll take three separate, messy data sources and turn them into one clear KPI system and reporting setup.
- Business Ethics - we'll be careful with customer data and business information the whole way through, keeping anything private or sensitive protected.

Methodology & Methods Selection

We're going to use a mix of methods to understand Bumby Wool's business and answer the questions above. Here's what each one means and why we're using it.

I. Descriptive Analytics

First, we'll just look at the numbers as they are today — no fancy modeling, just a clear picture of:

- total revenue and average order value
- which products and customizations are selling
- how much traffic the website is getting
- how much is being spent on marketing, and how far it's reaching

This gives us a starting point a snapshot of where the business stands right now.

II. Exploratory Data Analysis (EDA)

Next, we'll dig a bit deeper to spot patterns like whether spending more on ads actually leads to more traffic and sales, or whether certain months are consistently busier than others.

III. Funnel Analysis

A customer buying from Bumby Wool typically goes through these steps:

- Visits the website → Looks at a product → Adds it to cart → Goes to checkout → Buys it

We'll look at each step using Google Analytics and Shopify data to find out where people are dropping off, so we know exactly where fixing things would make the biggest difference.

IV. Customer Segmentation & Retention Analysis

We'll split customers into two groups - new buyers and repeat buyers and look at who's coming back. This helps show which products or marketing channels bring in one-time shoppers versus customers who stick around.

V. Marketing Performance Analysis

We'll check how well Meta ads are actually performing, using simple metrics like Return on Ad Spend (ROAS), Cost per Acquisition (CPA), and click-through rate. Then we'll compare that against real Shopify sales to see if the ads are truly paying off.

VI. Dashboard Development

Finally, we'll build a simple, interactive dashboard (in Power BI or Looker Studio) that shows the key numbers in one place — something a small team can actually keep updating without a headache.

Data Sources

Here's the data we'll be working with, either given to us by Bumby Wool or pulled from each platform's own reports:

- Shopify and WordPress - orders, products, customer purchase history, discounts, and refunds
- Meta Ads Manager - campaign details, ad spend, reach, clicks, and conversions
- Google Analytics (GA4) - where traffic comes from, sessions, and how people move through the site before buying

We won't include any personal customer details - no names, emails, phone numbers, or addresses anywhere in our analysis, dashboard, or report. If anything like that shows up in the data, we'll remove it or anonymize it before using it.

Tools and Technologies

Tool	Purpose
Microsoft Excel	Data cleaning & preparation
Python	Data analysis & KPI calculations
Power BI / Looker Studio	Dashboard creation & visualization
SQL (if required)	Data querying & filtering across sources
Microsoft Word	Report writing
Microsoft PowerPoint	Final presentation

High-Level Timeline

Week	Activity
Week 8	Understand datasets and prepare project plan
Week 9	Clean and organize Shopify, Meta, and GA data
Week 10	Perform exploratory data analysis and validate KPI framework
Week 11	Analyze marketing, revenue/product, and customer retention data
Week 12	Build Power BI / Looker Studio dashboard
Week 13	Develop business recommendations
Week 14	Prepare final report and presentation
Week 15	Review, improve, and submit final project

Appendix A — Project Gantt Chart

This chart shows our plan for Weeks 8 through 15 — what needs to get done, who's doing it, and roughly when. We'll use it to keep track of our progress and make sure we're not falling behind.

Task	Owner	8	9	10	11	12	13	14	15
Project Planning	All Members								
Data Collection & Access Setup	Baranitharan Balraju								
Data Cleaning	Marvelous Chidiebube Ihedigbo								
Data Integration (Shopify/Meta/GA)	Marvelous Chidiebube Ihedigbo								
Exploratory Data Analysis	Rahul								
Marketing (Meta) Analysis	Rahul								
Revenue & Product Analysis	Arun Saji								
Customer Retention Analysis	Antony Boon Clinu Mangatt Raju								
Dashboard Development	Arun Saji								
Dashboard Testing	Baranitharan Balraju & Antony Boon Clinu Mangatt Raju								
Business Recommendations	Rahul & Antony Boon Clinu Mangatt Raju								
Final Report	All Members								
Final Presentation	All Members								
Final Review	All Members								

Appendix B — Team Roles and Responsibilities Matrix

Team Member	Primary Responsibilities	Supporting Responsibilities
Baranitharan Balraju	Data Collection & KPI Framework Development	Dashboard Testing, Project Coordination & Final Review
Marvelous Chidiebube Ihedigbo	Data Cleaning & Data Integration (Shopify/Meta/GA)	Data Validation & Quality Assurance
Rahul	Marketing (Meta) Analysis & Business Recommendations	Exploratory Data Analysis Support & Report Editing
Arun Saji	Revenue/Product Analysis & Dashboard Development	Business Recommendations & Data Validation
Antony Boon Clinu Mangatt Raju	Customer Retention Analysis & Report Writing	Dashboard Testing & Client Communication
All Members	Project Planning, Final Report, Final Presentation & Final Review	Team discussions, quality assurance & client communication

Appendix C — Risk Assessment & Mitigation Strategies

Here's what could go wrong along the way, and what we'll do about it if it does.

Risk	Impact	Likelihood	Mitigation Strategy
Data from Shopify, Meta, or Google Analytics is missing or incomplete	High	Medium	Check all the data early on, note anything that's missing, and write down any assumptions we make along the way.
The client is slow to give us access or answer questions	Medium	Medium	Reach out to the client early when we need something, and keep working with whatever data we already have while we wait.
Tracking isn't consistent across platforms (like missing tags or tracking gaps)	High	Medium	Flag these gaps early in our current-state analysis and be upfront about any limits this puts on our results.
Data has duplicates or doesn't match up across sources	High	Medium	Clean and double-check the data before we start any real analysis.
Team members get busy with other coursework or personal stuff	Medium	Medium	Meet weekly, split up tasks fairly, and keep each other posted on progress.
Software, files, or data connectors stop working properly	Medium	Low	Keep backup copies of everything and have a manual backup plan (like exporting data by hand) if needed.
We fall behind schedule	High	Low	Stick to the Gantt chart, check in on progress weekly, and get key tasks done ahead of deadlines, not at the last minute.

Appendix D — Data Governance & Ethics Checklist

We'll handle Bumby Wool's data carefully and responsibly, keeping everything confidential the whole way through.

Data Governance Requirement	Compliance Status	Implementation
Customer names won't show up anywhere in our analysis or final report.	Compliant	We'll remove names or swap them for anonymous IDs before we start analyzing.
Phone numbers, emails, home addresses, and payment info will be removed or hidden.	Compliant	No personal identifying details will be included in any of our working data.
The data is only being used for this school project.	Compliant	We'll only use it to complete the MGMT 6219 Capstone Project — nothing else.
Only our team can access the project files.	Compliant	Files will be kept secure and only shared with the five of us.
Results will be shared as grouped, anonymous data — not individual records.	Compliant	No individual customer will be identifiable in any report, dashboard, or presentation.
The dashboard won't expose any private business or customer info.	Compliant	It'll only show summarized numbers, nothing sensitive or identifiable.
We'll be honest and accurate with the analysis.	Compliant	We'll report what the data actually shows, without twisting the results.
We'll follow the ethical principles we've learned in our Business Ethics coursework.	Compliant	We'll stick to standards around privacy, confidentiality, and responsible data use.